DO NOW — RECALL

Lesson 128 · Run-Length Encoding: Encoding and Decod

Name: _____ Class: _____ Date: _____

1. When must lossless compression be used?

2. In a queue, what does FIFO stand for?

3. What is a function parameter?

4. In a BST, which side holds a smaller value?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

When exact reconstruction is necessary, such as executable code.

2. Model answer

First in, first out.

3. Model answer

An input used by a subroutine.

4. Model answer

The left subtree.

By the end, you can…

• Group identical adjacent symbols, not identical symbols separated by others.

• Record every run as count and symbol consistently.

• Count storage using an explicit format, including counts and delimiters.

Vocabulary — know these terms

RLE — Encodes consecutive equal symbols as a symbol and count.

run — A consecutive sequence of identical symbols.

overhead — Extra data required by an encoding scheme.

lossless — Decoding reproduces the original exactly.

NEW LEARNING

Read, annotate and explain

Identify runs

Group identical adjacent symbols, not identical symbols separated by others.

Encode

Record every run as count and symbol consistently.

Check cost

Count storage using an explicit format, including counts and delimiters.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Encode AAAAABBCCC using (count,symbol) pairs; verify the result.

AAAAA | BB | CCC

Predict the first step from the prompt.

(5,A)(2,B)(3,C)

Explain the intermediate result or decision.

Decode original

AAAAA =5A, BB=2B, CCC=3C. Encoded result (5,A)(2,B)(3,C); decoded data restores AAAAABBCCC exactly.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply RLE to AAAABBBCCAAAA using count-symbol pairs. [6]

Question 2

Decode (3,X)(1,Y)(2,X) and explain why runs must retain order. [5]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A bitmap row has 40 alternating pixels. Explain why RLE might increase file size and identify a better-suited pattern. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

RLE always reduces the size of any input.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.